Tomato Leaf Disease Detection using YOLOv8

Dataset Link: Tomato Leaf Disease Detection – YOLOv8 Dataset

1. Domain Background

Tomatoes are among the most widely used vegetables globally and are cultivated almost everywhere. Unfortunately, according to Singh, Singh, and Kumar (2017), tomatoes are highly susceptible to diseases, especially during the cultivation process and post-harvest. More than 200 diseases can impact plant health, leading farmers to use pesticides that lower the quality of the produce and risk spreading infections to nearby plants. These issues contribute to reduced agricultural yield and economic loss.

Recent advances in computer vision and deep learning offer promising solutions. They enable quick and accurate detection of plant diseases, potentially saving crops. Models like YOLO (You Only Look Once) are particularly effective for detecting diseases in small objects such as plant leaves.

2. Problem Statement

Detecting plant diseases manually is time-consuming and error-prone. Due to high tomato demand, plantations are often large, making manual inspection inefficient. Pre-trained computer vision models can significantly improve accuracy and efficiency while reducing labor costs. The goal of this project is to develop a robust real-time model capable of classifying 10 categories of tomato leaf diseases using YOLOv8.

3. Solution Statement

This project proposes training the latest YOLOv8 model for precise object detection. The model will be trained on a large tomato leaf dataset from Kaggle and will classify tomato leaf health in near real-time. The final model is expected to be compatible with mobile devices, enabling field usage.

Project Steps:

- Data preprocessing
- YOLOv8 fine-tuning

- Performance evaluation on test set
- Result comparison with baseline models

4. Inputs and Datasets

The project uses the **Tomato Leaf Disease Detection – YOLOv8 Dataset** from Kaggle, which includes 10,853 labeled images across 10 classes:

- Tomato Bacterial Spot
- Tomato Early Blight
- Tomato Late Blight
- Tomato Leaf Mold
- Tomato Septoria Leaf Spot
- Tomato Spider Mites
- Tomato Target Spot
- Tomato Yellow Leaf Curl Virus
- Tomato Mosaic Virus
- Tomato Healthy

Dataset Structure:

- Training Set: 7,842 images (72%)
- Validation Set: 1,960 images (18%)
- Test Set: 1,051 images (10%)

Image Resolution: 640x640
Preprocessing Applied:

- Resizing and reorienting images
- No additional augmentations

Annotations: YOLOv8 Format

5. Baseline/Benchmark Models

Two benchmark models will be used for comparison:

- YOLOv5 with the same image resolution
- MobileNetV2 (adapted for object detection via bounding boxes)

These models will be trained without fine-tuning on the same dataset to serve as benchmarks in terms of accuracy, precision, recall, and inference time.

6. Evaluation Metrics

- **Recall and Precision:** Accuracy per class
- **F1 Score:** Harmonic mean of precision and recall
- **mAP (mean Average Precision):** Standard object detection metric
- **Time:** Real-time capability evaluation

Metrics will be applied to both validation and test sets for generalization.

7. Project Design

Prepare the Dataset

- Data cleaning and verification (remove corrupted images and duplicates)
- Apply data augmentation (brightness, zoom, flipping, rotation)
- Organize data to match YOLOv8 format

Train the Model

- Use YOLOv8 model pre-trained on COCO dataset
- Fine-tune with tomato leaf data
- Tune hyperparameters (learning rate, batch size, epochs)

Evaluate the Model

- Use mAP and F1 Score
- Create visualizations for model predictions

Baseline Model Evaluation

- Train YOLOv5 and MobileNetV2
- Compare metrics with YOLOv8

8. Frameworks and Tools

- Python, PyTorch, Ultralytics YOLOv8
- OpenCV
- TensorBoard